

RAJVANSH MALHOTRA

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EDUCATION

National Institute of Technology, Kurukshetra

B.Tech, Computer Engineering

Aug 2023 – Jun 2027

PROFESSIONAL EXPERIENCE

Samvedya AI, in collaboration with AIIMS Delhi

Mar 2025 – May 2025

AI/ML Intern

- **Developed a LLM Pipeline** to convert PNG-based digital designs into SVG format, integrating image analysis, LLM orchestration, and API pipelines to automate a visual abstraction workflow.
- Delivered a functional MVP within 3 weeks using Python and modern ML techniques, streamlining visual abstraction for clinical research.
- Engineered a **RAG pipeline** to generate context-aware answers in a chat-based system, achieving **80% accuracy** on clinically relevant visual abstracts at AIIMS Delhi.

National Institute of Technology, Kurukshetra

Jun 2024 – Dec 2024

Machine Learning Research Intern

- Engineered features through **extracting image features with VGG19 in TensorFlow and selecting top 1,000 impactful features via a Random Forest classifier**.
- **Improved model accuracy from 60% to 82% by training a Vision Transformer** on the dataset.
- Optimised predictions by integrating Golden Ratio + eye-spacing calculations via OpenCV.
- Achieved state of the art accuracy with a custom ANN on structured data, reducing training time and inference costs.

PROJECTS

Battery Health Optimization

Mar 2026

Research Work – IEEE

- Deployed a **Physics-Informed Residual Neural Network (RPNN)** on an **ESP32 microcontroller** for real-time **SOC and SOH estimation**, embedding Coulomb counting differential equations directly into the training loss to enforce physically consistent predictions.
- Trained with a three-component loss combining **data-driven MSE, ODE physics residual via automatic differentiation**, and boundary constraints enforcing SOC within [0, 100]%, optimized using **AdamW** with physics loss weight $\lambda = 0.1$.
- Compressed the PyTorch RPNN into a **C++ TinyML engine for edge deployment**, eliminating cloud dependency for safety-critical battery diagnostics in **automotive applications**.

Gotham Orbital

Feb 2026

Agentic Satellite Intelligence Platform

- Engineered **optimum_ReAct**, a **custom agentic AI library from scratch**; deployed **3 concurrent LLM agents** (ATLAS, ORBITAL-1, ANALYST-1) with isolated context windows and role-specific system prompts, inspired by Palantir's Gotham.
- Designed **HybridMemory** combining **BM25 retrieval, sentence-transformer semantic search**, and a weighted knowledge graph with multi-factor eviction (importance, recency decay, access frequency, connection weight).
- Deployed **production backend on AWS EC2** with Docker containerisation, async Semaphore concurrency controls, and **exponential backoff retry (15/30/60s)** for Groq API fault tolerance.
- Shipped end-to-end **MLOps pipeline via GitHub Actions CI/CD**.

Rizzler

Jun 2025

LLM Fine-Tuning Project

- Fine-tuned **Meta LLaMA-3 8B** using **QLoRA (Quantized LoRA)** via Unsloth for efficient low-resource conversational training.
- Applied **LoRA adapters** to key transformer components (**q_proj, v_proj**) with **r=64 and 4-bit quantisation** to reduce memory usage while preserving performance.
- Tuned generation outputs using **temperature and top_p sampling**.

TECHNICAL SKILLS

Languages	Python, C/C++
ML / AI	Machine Learning, Deep Learning, NLP, Large Language Models, LLM Fine-tuning, RAG, Prompt Engineering, Scikit-learn, TensorFlow, PyTorch, Pandas, Matplotlib, OpenCV
DevOps	AWS EC2, Docker, GitHub Actions, MLflow, Apache Airflow, ETL, SQL
Coursework	Data Structures & Algorithms, OS, DBMS, Computer Networks, Computer Architecture, Software Engineering, Artificial Intelligence

ACHIEVEMENTS

- **Amazon ML Challenge:** Directed a high-performing team to the top **97.2 percentile**; reduced SMAPE from 100 to 48 within 2 days using targeted ML strategies.
- **Problem Solving:** Solved 300+ problems on LeetCode and other coding platforms.
- **Achieved 1 silver medal and 3 gold medals** in Taekwondo .